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ACADEMIC VAULT

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PHA 101: Introduction to Pharmacy - Drugs, Dosage Forms and the Pharmacist's Role

Orientation to pharmacy practice, drug nomenclature and pharmaceutical calculations

FACULTY PRIMER

Faculty Primer

Faculty of Pharmaceutical Sciences
Pharmaceutics & Pharmaceutical Technology | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack orients new pharmacy students to the profession, explains drug nomenclature and dosage forms, and drills the pharmaceutical calculations that catch students out in early assessments, including percentage concentrations, dilutions and dose calculations.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the pharmacist's roles in dispensing, counselling, compounding and pharmacovigilance.
- Distinguish chemical, generic and brand names and discuss bioequivalence.
- Classify dosage forms and explain how the route of administration affects onset and bioavailability.
- Perform percentage, dilution and dose calculations accurately with correct units.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What pharmacy is for

Pharmacy is the science and practice of preparing and using medicines safely and effectively.

- Dispensing: verifying the prescription, preparing and labelling the medicine.
- Counselling: explaining dose, frequency, duration, storage and side effects to the patient.
- Compounding: extemporaneous preparation where no commercial product fits.
- Pharmacovigilance: detecting, assessing and reporting adverse drug reactions.
- Regulation: controlled drugs, prescription requirements and record keeping.
- Medication reconciliation and adherence support reduce avoidable harm.

> **The pharmacist's check**
> Right patient, right drug, right dose, right route, right time and right documentation. Any doubt is a reason to stop and verify, not to guess.

Names, dosage forms and bioavailability

Names and formulations are not trivia; they determine how a medicine behaves in the body.

- Chemical name describes structure; generic name is the internationally accepted non-proprietary name; brand name is the manufacturer's trade name.
- Bioavailability is the fraction of an administered dose that reaches systemic circulation unchanged.
- Bioequivalence means two products deliver the same rate and extent of absorption within accepted limits.
- Oral solid forms: tablets, capsules, granules. Modified-release designs alter release rate.
- Liquid forms: solutions, suspensions, emulsions; shake suspensions before measuring.
- Parenteral, topical, rectal, inhaled and ophthalmic routes each have distinct advantages and risks.

Calculations without panic

Set every calculation out in one consistent format: known, required, conversion, equation, answer with units.

- Percentage weight in volume and weight in weight are not interchangeable; read what the label states.
- Dilution: C1 x V1 = C2 x V2, with concentrations in the same unit.
- Dose by body weight: dose in mg per kg multiplied by weight in kg, then checked against the maximum adult dose.
- Dose by body surface area requires the stated formula and correctly converted height and weight.
- Alligation solves mixing problems between two strengths.
- Always sanity-check: an answer that implies 40 tablets per dose is almost certainly wrong.

> **Units decide safety**
> Milligrams and micrograms differ by a factor of 1000. A decimal slip can be fatal. Convert to the same unit before calculating and write the unit at every step.

From molecule to medicine

A drug is a chemical producing a biological effect; a medicine is a formulated product that delivers it safely. The difference is pharmaceutical science: solubility, stability, particle size, excipients, packaging and storage determine whether the active ingredient reaches the patient in usable form.

Pharmacokinetics describes what the body does to the drug - absorption, distribution, metabolism, excretion. Pharmacodynamics describes what the drug does to the body. Most clinical questions about dosing are kinetics questions in disguise.